

[← Go Back](#)

Hardware

Nicla Sense ME 

Tutorials

Arduino Nicla Sense ME
Cheat Sheet

Sensors Readings on a
Local Webserver

Connecting the Nicla
Sense ME to IoT Cloud

Getting Started with Nicla
Sense ME

Arduino Nicla Sense ME as
a MKR Shield

Nicla Sense ME User
Manual

Displaying on-Board Sensor
Values on a WebBLE
Dashboard

Home / Hardware / Nicla Sense ME /
Connecting the Nicla Sense ME to IoT
Cloud

Connecting the Nicla Sense ME to IoT Cloud

This tutorial shows you how to set up the
Arduino Nicla Sense ME and the Arduino
Portenta H7 to connect to Arduino Cloud
IoT and upload sensor data.

Author	Ernesto	Last revision	· 09/17/2025
	Lopez,		
	Pablo		
	Marquínez		

ON THIS PAGE

Overview

- Goals +
- Instructions +
- Conclusion +
- Troubleshooting +

Overview

In this tutorial you will learn how to upload data from the Nicla module to the IoT Cloud. You will use the **Portenta H7** to interface with the **Nicla Sense ME** using the ESLOV connector and upload the data using the Portenta Wireless capabilities.

Goals

How to configure the Portenta H7 to read temperature values from the Nicla Sense ME using the ESLOV connector.

How to Connect the Portenta H7 to the Arduino Cloud

Publish the temperature values that you obtained from the Nicla board to the Arduino Cloud.

Required Hardware and Software

Portenta H7

Nicla Sense ME

ESLOV cable (included with the Nicla Sense ME)

USB-C® to USB-A / C
depending on your hardware
(Portenta H7)

USB-A to micro USB-A (Nicla Sense ME)

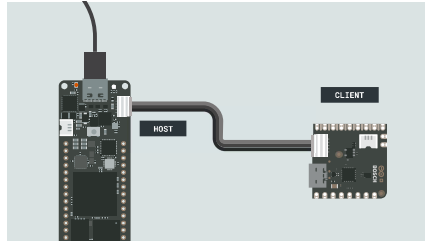
Wi-Fi Access point with access to the internet

Arduino Cloud account

Instructions

Hardware Connection

For the hardware setup, just connect the Nicla board to the Portenta H7 using the ESLOV cable like in the illustration below. Then connect the Portenta H7 to your computer using an USB-C® cable.



Setup Eslov Communication with the NICLA Board

There are three ways to read from the on-board sensors:

- 1 Read the sensors directly from Nicla Sense ME in standalone mode.
- 2 Read sensor values through Bluetooth® Low Energy
- 3 Read sensor values through UART by connecting an ESLOV cable.

For further tips on how to operate the Nicla module check the [cheat sheet](#).

Client Board (Nicla Sense ME)

The **Nicla Sense ME** will be listening to the Host board to send back the required data, this is all automated via the libraries **Arduino BHY2** and

The code is available inside the examples provided with the **Arduino_BHY2** Library. Open it by going to **Examples > Arduino_BHY2 > App.ino**

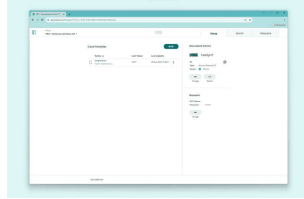
This is the code, which initialize the sensors, and maintain the communication:

```
1  /*
2   * Use this sketch if yo
3   * an external device ac
4   * Here, nicla just reac
5   * the eslov port or thr
6   */
7
8  #include "Arduino.h"
9  #include "Arduino_BHY2.h"
10
11 void setup(){
12     BHY2.begin(NICLA_I2C
13 }
14
15 void loop(){
16     // Update and then s
17     BHY2.update(100);
18 }
```

Set up the Arduino Cloud

To configure the Arduino Cloud you can follow the tutorial [Getting Started with the Arduino Cloud](#).

Create a new Thing at <https://create.arduino.cc/iot/things>, you can call it "PRO - Portenta and Nicla". You will need to attach the Portenta H7 as a new device to your **Thing setup**. After that, go to **Variables**, click the **add** button and select a float variable called **temperature** to store the temperature readings.



Arduino Cloud - Thing Setup



Remember to add your Wi-Fi SSID name and its password (you can do this inside the Devices tab on the device page) to be able to connect to the Arduino Cloud. This is also done when connecting your board to the Cloud

Host Board: Edit the Cloud Sketch

You can edit the sketch by clicking the **sketch tab** inside your **Thing page**. The sketch is automatically generated with enough code to upload it and connect it to the Cloud.

Before uploading, you should add the following code:

First include the headers that you need and declare the temperature sensor by adding the **temperature sensor ID**.

```
1 #include "thingProperties.h"
2 #include "Arduino.h"
3 #include "Arduino_BHY2Host.h"
4
5 Sensor tempSensor(SENSOR_ID)
```

You can find all the Sensor IDs at

 <https://docs.arduino.cc/tutorials/nicla-sense-me/cheat-sheet#sensor-ids>.

Inside `void setup()` initialize the `Serial` communication, set up the variables and configuration for the Arduino Cloud (properties) and wait until the **Portenta H7** is connected to the Wi-Fi and IoT Cloud. At this point, the communication with the **Nicla Sense ME** will be set up and you can start configuring the temperature sensor.

Note: Now we are using "NICLA_VIA_ESLOV". In case you mount it as a shield use

 "NICLA_AS_SHIELD" as the second parameter of the `begin()` function, or "NICLA_VIA_BLE" if you use Bluetooth® Low Energy.

```

1 void setup(){
2   Serial.begin(9600);
3   delay(1500);
4
5   Serial.print("Confi
6   // Defined in thing
7   initProperties();
8
9   // Connect to Ardui
10  ArduinoCloud.begin(
11
12  // Wait to be conne
13  Serial.println("Cor
14  while (ArduinoCloud
15    ArduinoCloud.upda
16    delay(500);
17  }
18
19  delay(1500);
20
21  Serial.println("Ini
22  BHY2Host.begin(fals
23
24  //If you want to co
25  //while(!BHY2Host.k
26
27  tempSensor.configur
28  temperature = tempS

```

If you use

```
yourSensor.begin(),
```

it will be configured the same as

i with

```
yourSensor.configure(1
,0)
```

If the Nicla Sense ME communicates through Bluetooth® Low Energy, we recommend wrapping

```
BHY2Host.begin(false,
NICLA_VIA_BLE)
```

in a **while** clause to make sure the connection is established before the sketch continues.

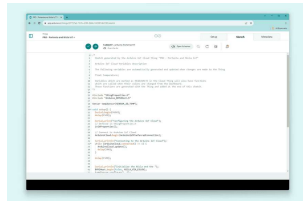
Inside the **void loop()** function you will make the **Portenta H7** get all the needed data from the **Nicla**

update the data to the Arduino Cloud.

```
1 void loop(){
2   BHY2Host.update();
3   temperature = tempSens
4
5   Serial.print("Temperatu
6   Serial.println(temperat
7
8   ArduinoCloud.update();
9 }
```

Upload the sketch from the **sketch tab** by clicking the second button at the top left side of the sketch bar.

Once it has been uploaded, you can see the temperature value just uploaded by going to your **Thing Setup** tab and looking at the last value of the **temperature** variable. You can also open the Serial Monitor to see your data live.



Arduino Cloud - Sketch
tab

The Sketch:


```
1  /*
2   Sketch generated by t
3
4   Arduino Cloud Variabl
5
6   The following variabl
7
8   float temperature;
9
10  Variables which are n
11  which are called wher
12  These functions are c
13  */
14
15  #include "thingPropert:
16  #include "Arduino_BHY2I
17
18  Sensor tempSensor(SENSC
19
20  void setup() {
21    Serial.begin(9600);
22    delay(1500);
23
24    Serial.print("Configu
25    // Defined in thingPr
26    initProperties();
27
28    // Connect to Arduinc
```

Conclusion

In this tutorial you learned how to upload the temperature values from the Nicla Sense ME to the Arduino Cloud. You followed the process to set up the Nicla board in order to send data via ESLOV connection and you configured the Arduino Cloud to receive the temperature data.

Next Steps

Try to upload other sensor data from the Nicla Sense ME. You can see the available sensors in the [cheat sheet](#).

Experiment with the

for a more sophisticated project.

Troubleshooting

Arduino Cloud

If you encounter any issue in the process of using the Arduino Cloud, please visit the [Getting Started with Arduino Cloud](#)

Suggest changes	Need support?	License
The content on docs.arduino.cc is facilitated through a public GitHub repository. If you see anything wrong, you can edit this page here .	Help Center Ask the Arduino Forum Discover Arduino Discord	The Arduino documentation is licensed under the Creative Commons Attribution-Share Alike 4.0 license.

Was this article helpful?

